

Experiment 11

11.1.1 Sum of digits

Algorithm :

Step 1 : Start

Step 2 : Input an integer and store it in num

Step 3 : Set sum to 0

Step 4 : Repeat steps 5 through 7 while num is greater than 0

Step 5 : Find the last digit using the modulo operator:

$\text{digit} = \text{num} \% 10$

Step 6 : Add the digit to the current sum

Step 7 : Remove the last digit using integer division:

$\text{num} = \text{num} // 10$

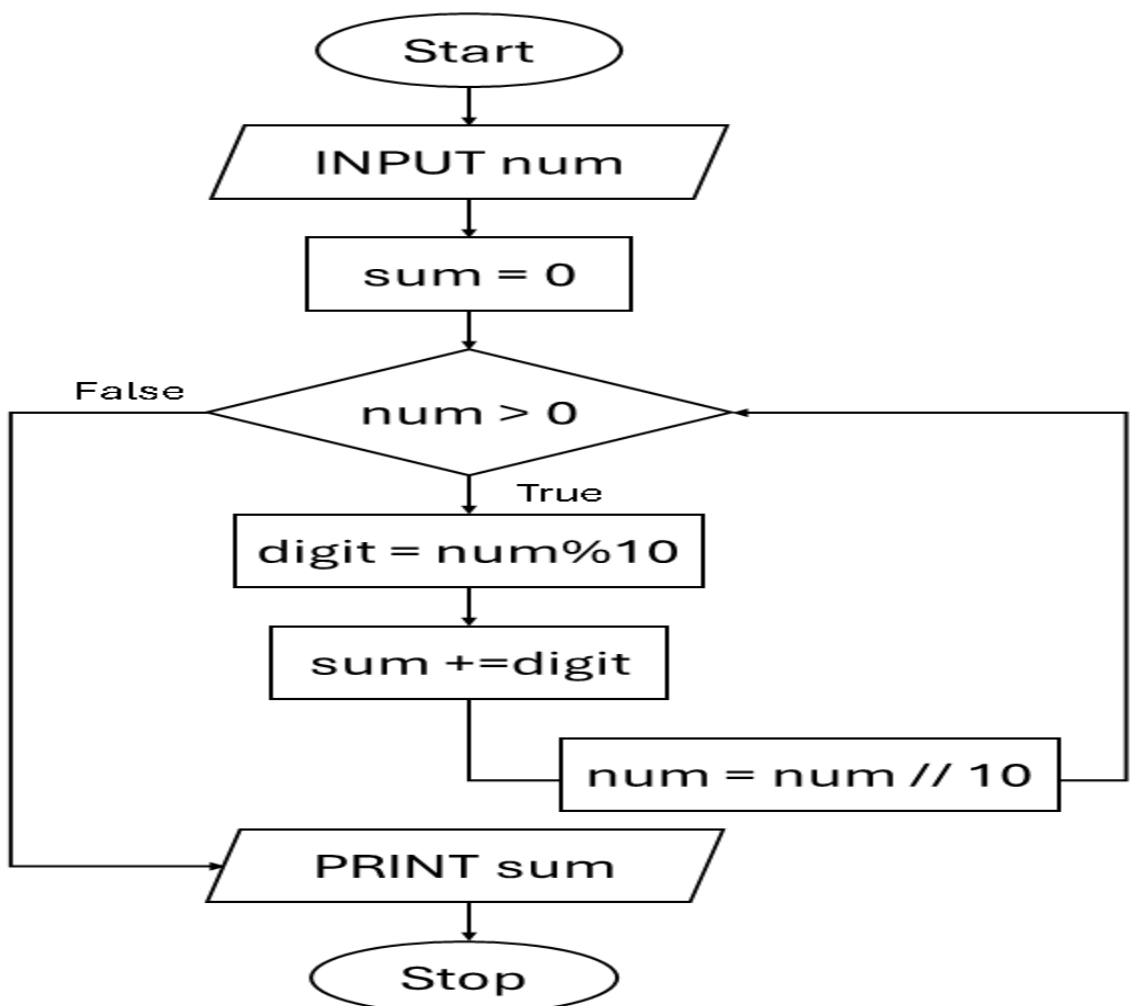
Step 8 : Go back to Step 4

Step 9 : Once num becomes 0, exit the loop

Step 10 : Print the final value of sum

Step 11 : Stop

Flowchart :



```
num = int(input("Enter a number: "))
sum=0
while num>0:
    digit=num%10
    sum+=digit
    num=num//10
print("Sum of digits:",sum)
```

Execution :

08:15

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11.1.1.1. Sum of Digits of a Number

Write a Python program that reads an integer, calculates the sum of its digits, and prints the result.

Input Format:

- The program reads an integer input with the prompt "Enter a number: "

Output Format:

- Print the sum of the digits in the format:

Sum of digits: <digit_sum>

Note:

- Some base code was provided (input and print statements). The task was to complete the missing logic.

Sample Test Cases

Test case 1

Enter a number: 123

Sum of digits: 6

Debugger

sumofdig...

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1 num = int(input("Enter a number: "))

2 sum=0

3 while num>0:

4 digit=num%10

5 sum+=digit

6 num=num//10

7 print("Sum of digits:",sum)

Average time

0.006 s

5.75 ms

Maximum time

0.008 s

8.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8ms

Test case 2 4ms

Terminal

Test cases

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